

Banking Data Engineering Pipeline Project Report

Platform: Databricks Free Edition

1. Project Objective

Design and implement an end-to-end banking data engineering pipeline using Databricks Free Edition, PySpark and Delta Lake. The pipeline is modular, dynamic and capable of processing multiple banking transaction sources.

2. Technologies

Databricks Free Edition, Unity Catalog Volumes, PySpark, Spark SQL, Delta Lake, Python, Git/GitHub.

3. Current Architecture

Source Systems (ATM, UPI, NEFT, RTGS, CARD, INTERNET_BANKING, MOBILE_BANKING, BRANCH)

- incoming
- 01_Ingestion
- Bronze
- 02_Data_Validation
- Silver
- Gold
- SQL Analytics & Dashboard

4. Folder Structure

incoming/, reference_data/, processed/, rejected/, archive/, bronze/, silver/, gold/, checkpoints/, logs/, metadata/, temp/

5. Completed Work

- 00_Setup_Folders
 - Dynamic creation of all project folders.
 - Dynamic creation of banking source folders.
 - Idempotent execution (safe to run multiple times).
- 00_Precheck
 - Validation of project folders.
 - Dynamic discovery of source systems.
 - Validation of landing folders.
 - Pipeline stops if required folders are missing.
- 01_Ingestion
 - Automatically discovers source folders.
 - Reads all CSV files dynamically.
 - Applies predefined schema.
 - Adds audit columns (source_system, source_file, pipeline_name, pipeline_run_id, load_date, load_timestamp).
 - Writes data into Bronze Delta layer.
 - Moves processed files into processed/.
 - Prints execution summary.

6. Dynamic Features

- Dynamic source discovery
- Dynamic CSV detection
- Reusable folder structure
- Delta Lake storage
- Audit columns
- Modular notebook design
- Compatible with Databricks Free Edition

7. Remaining Development

02_Data_Validation
03_Bronze_to_Silver
04_Customer_Master
05_Fraud_Detection
06_Gold_Layer
07_Audit_Logging
08_SQL_Analytics
09_Dashboard
10_GitHub Documentation

8. Data Validation (Next Phase)

Schema validation
Mandatory column validation
Duplicate detection
Null value checks
Negative amount validation
Invalid timestamp validation
Rejected records handling

9. Fraud Detection (Planned)

High-value transactions
Rapid repeated transactions
Multiple branches within short interval
Blacklisted customer detection
Duplicate transaction detection

10. Future Enhancements

Production migration to Azure Databricks:

- Auto Loader
- ADLS Gen2
- Azure Data Factory
- Delta Live Tables
- Scheduled Workflows

11. Resume Highlights

Designed a dynamic end-to-end banking data engineering pipeline using Databricks Free Edition, PySpark and Delta Lake. Implemented Medallion Architecture, dynamic ingestion, reusable validations, audit logging foundation and scalable project structure.